

Abstract

Lysosomal alpha-mannosidase is an exoglycosidase responsible for the sequential degradation of complex, hybrid, and high-mannose N-linked oligosaccharides. The absence or deficiency of this enzyme leads to the development of alpha-mannosidosis, a hereditary lysosomal storage disorder. Consequently, this enzyme holds significant importance in the pharmaceutical industry, which currently represents its sole application.

The primary objective of this study was the development of an industrial plant for the production and purification of recombinant human lysosomal alpha-mannosidase, based on patent AU2011220163B2. To achieve this, it was necessary to evaluate the enzyme market, describe the production processes required to yield the desired final product, size relevant process equipment, and perform an economic analysis.

The market for this enzyme is experiencing rapid growth and was valued at €17.3 billion in 2024. When evaluated specifically for the treatment of this disorder, the market is valued at €240.9 million, with a projected Compound Annual Growth Rate (CAGR) of 10% over the next five years. Assuming a 100% market penetration for this project, an annual drug production of 2.3 kg was estimated to meet market demand. This drug is commercialized in vials containing 10 mg of alpha-mannosidase.

Enzyme production is carried out via a fed-batch bioreaction using CHO cells, and its extraction and purification are achieved through a series of upstream and downstream processes, which are presented and described herein. Seven key pieces of equipment were sized: a production bioreactor, an air compressor, a pumping system, a heat exchanger, a storage tank, a centrifuge, and a chiller.

Furthermore, this project includes an economic analysis detailing the initial capital investment costs (€8.0 million), annual production costs (€15.9 million/year), and projected drug sales revenues (€975.00/vial, corresponding to €97.50/mg of alpha-mannosidase). The results substantiate the economic viability of establishing the designed industrial plant, which demonstrates high profit margins and substantial cash flow.

Additionally, the study presents a proposed plant layout, alongside sensitivity, SWOT, and safety analyses. Finally, future recommendations are provided to optimize the process and ensure a more realistic and sustainable approach.

Process Design

The created flow diagram of the production process and industrial purification of alpha-mannosidase was based on the patent AU2011220163B2.

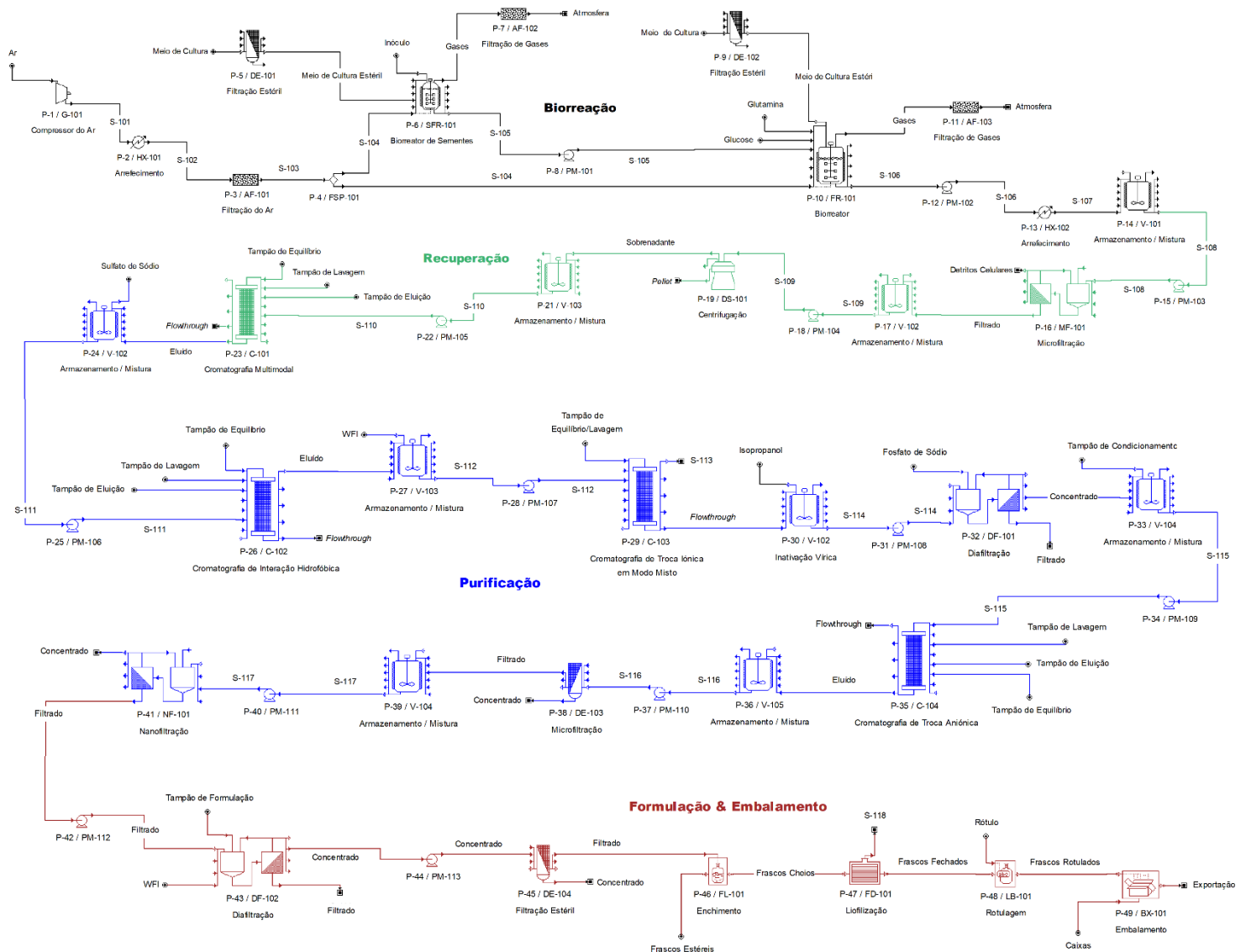


Figure 1. Diagram of the production and purification process of human alpha-mannosidase.

*The project was done in Portuguese, thus the text present in pictures is in Portuguese.